



DEPARTMENT OF AIML & DATA SCIENCE

INTRODUCTION TO NATURAL LANGUAGE PROCESSING (AI532P)

Academic Message Classifier using NLP

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Abstract

This project presents an Academic Message Classifier that automatically categorizes unstructured academic-related messages such as Events, Tech Talks, Competitions/Auditions, and Other Academic notices. The objective is to simplify how students handle important academic updates by extracting key details like dates, times, and deadlines, and by simulating reminders for actionable items.

The dataset used consists of 50 sample academic messages distributed across four categories. Preprocessing included text cleaning, tokenization, and stopword removal. TF-IDF was applied for feature extraction, and a Multinomial Naïve Bayes classifier was trained to predict categories. OCR (pytesseract) was integrated to handle scanned images, while regex-based rules were used to extract event-specific details.

Results demonstrated accurate categorization of sample messages and successful extraction of structured details from both text and OCR inputs. For example, “*Cyber Security Workshop on 15th Sep 2025 at 11:30 AM*” was classified as a Tech Talk, with date and time correctly extracted, and a reminder successfully simulated. The system, while lightweight and effective, can be further improved with larger datasets and advanced models such as BERT for better contextual understanding.

Table of Contents

Sl.No.	Content	Page No.
1.	Introduction	4
2.	System Analysis	4
3.	Objectives	4
4.	System Design	5
5.	Methodology	6
6.	Implementation	6
7.	Results and Discussion	17
8.	Conclusion	18
9.	Reference	18

1. Introduction

Background

Academic institutions generate frequent announcements related to events, workshops, competitions, and academics. Students often find it challenging to track these messages when they come in unstructured formats like text messages, WhatsApp forwards, or scanned images.

Need for NLP in this Problem

Traditional sentiment or keyword-based search methods fail to capture the context of academic messages. NLP methods like TF-IDF and classification models can effectively categorize and extract structured details from text, saving time and reducing missed deadlines.

Problem Definition

The problem is to automatically classify academic-related messages into meaningful categories and extract details such as event date, time, and registration deadlines.

Scope of the Project

This project demonstrates a basic NLP-based academic classifier. It includes OCR and regex-based information extraction. Future expansion includes handling larger datasets, adding contextual deep learning models, and integrating with calendar APIs for live reminders.

2. System Analysis

Problem Statement

Unstructured academic messages can overwhelm students and faculties, leading to missed deadlines and overlooked events. This project proposes a classifier to categorize such messages and provide reminders for actionable updates.

3. Objectives

- Build an NLP classifier for academic messages.
- Train a Multinomial Naïve Bayes model on TF-IDF features.
- Integrate OCR to extract text from image-based announcements.
- Extract event-specific details using regex.
- Simulate reminders for categories such as Events, Tech Talks, and Auditions.

4. System Design

The system architecture consists of:

1. **Input** – Accepts raw input in the form of text messages or images.
2. **Preprocessing** – Cleans the input text using tokenization, stopwords removal, and regex filtering.
3. **Feature Extraction (TF-IDF)** – Converts text into numerical feature vectors.
4. **Naïve Bayes Classifier** – Trains on TF-IDF features to predict the message category.
5. **Category Prediction** – Outputs the classified label (Event / Tech Talk / Audition / Other Academic).
6. **Regex Extraction** – Extracts structured information such as dates, times, and deadlines.
7. **Reminder Simulation** – Generates reminders for actionable categories like Event, Tech Talk, and Audition.

Module Description

- **Data Module** – Stores academic messages and their categories.
- **Preprocessing Module** – Cleans text, tokenizes, removes noise, and prepares data.
- **Feature Extraction Module** – Applies TF-IDF to represent text numerically.
- **Classification Module** – Uses Multinomial Naïve Bayes to predict message categories.
- **OCR Module** – Extracts text from scanned images using pytesseract.
- **Regex Extraction Module** – Detects event-related details such as date, time, and deadlines.
- **Reminder Module** – Simulates reminders for important academic activities.

5.Methodology

1. **Data Collection** – 50 academic messages across 4 categories.
2. **Preprocessing** – Cleaning, regex filtering, stopwords removal.
3. **Feature Extraction** – TF-IDF Vectorizer.
4. **Model Building** – Multinomial Naïve Bayes.
5. **OCR Integration** – Extract text using pytesseract.
6. **Regex Extraction** – Detects dates, times, deadlines.
7. **Evaluation** – Qualitative testing with default, user, and OCR inputs.

Libraries Used

- **pandas** – Data handling and dataset preparation.
- **scikit-learn** – Machine learning algorithms (TF-IDF, Naïve Bayes, train/test split).
- **pytesseract** – Optical Character Recognition (OCR) for extracting text from images.
- **Pillow (PIL)** – Image handling and preprocessing for OCR.
- **regex (re)** – Regular expression matching for cleaning text and extracting dates, times, and deadlines.

6. Implementation

Technology Stack

- Programming Language: Python
- Tools: Colab
- Libraries: pandas, sklearn, pytesseract, regex

CODE:

```
import pandas as pd

import re

import pytesseract

from PIL import Image

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import MultinomialNB


data = {

    "Message": [

        "Onam Celebration in College Auditorium tomorrow 10 AM",

        "Annual Sports Day next Monday at 9 AM",

        "Cultural Fest registrations open from 10th Sep",

        "Hackathon event this weekend",

        "Independence Day celebration on 15th Aug at 8 AM",

        "Farewell party for final years on 30th March 6 PM",

        "College Day function scheduled on 5th April 10 AM",

        "Blood Donation Camp in Seminar Hall on 20th July",

        "Alumni Meet on 25th December at 11 AM",

        "Christmas Celebration at College Auditorium on 23rd Dec",

        "Sports Meet closing ceremony on 28th Jan 3 PM",

        "Republic Day parade on 26th Jan at 8:30 AM",

        "Join AI Tech Talk on 10th Sep at 3 PM",

        "Web Development Workshop on Friday 5 PM",
```

"Data Science Seminar via Zoom on 12th Sep",
"Cyber Security Workshop on 15th Sep 2025 at 11:30 AM",
"Cloud Computing Seminar on 2nd Nov 10 AM",
"Android App Development Workshop on 18th Feb at 2 PM",
"AI Ethics Webinar on 25th March at 4 PM",
"Robotics Seminar on 7th July at 10:30 AM",
"IoT Workshop scheduled on 10th Aug at 9 AM",
"Blockchain Seminar on 15th Oct 11 AM",
"Machine Learning Guest Lecture on 5th Dec 3 PM",
"Big Data Analytics Workshop on 20th Jan 2 PM",

"Auditions for College Drama on 15th Sep 2 PM",
"Coding Competition registration opens 12th Sep",
"Music Competition on 18th Sep at 6 PM",
"Dance Auditions on 22nd Nov at 4 PM",
"Quiz Competition on 1st Dec 10 AM",
"Poster Making Competition on 12th Feb 9 AM",
"Hackathon registrations open till 25th March",
"Photography Contest submission deadline 30th June",
"Debate Competition on 10th July 11 AM",
"Poetry Recital on 15th August 3 PM",
"Drama Club Auditions on 5th Oct 2 PM",
"Gaming Tournament on 18th Dec at 5 PM",

"Assignment 2 submission deadline is tomorrow",
"Library closed today due to maintenance",
"Exam timetable released for 3rd semester",

"Submit assignment 3 by next Wednesday",
"Midterm exam schedule published",
"Final exam hall tickets available online",
"Project report submission deadline 10th April",
"Revaluation applications open till 25th May",
"Practical exam dates announced for 2nd year",
"Lab manuals submission last date 1st March",
"Holiday declared on 2nd October for Gandhi Jayanti",
"Online class links shared in student portal",
"Result publication for 4th semester on 15th June",
"Attendance shortage list released for verification"
],
"Label": [

"Event","Event","Event","Event","Event","Event",
"Event","Event","Event","Event","Event","Event",

"Tech Talk","Tech Talk","Tech Talk","Tech Talk",
"Tech Talk","Tech Talk","Tech Talk","Tech Talk",
"Tech Talk","Tech Talk","Tech Talk","Tech Talk",

"Audition","Audition","Audition","Audition",
"Audition","Audition","Audition","Audition",
"Audition","Audition","Audition","Audition",

```

        "Other Academic","Other Academic","Other Academic","Other Academic",
        "Other Academic","Other Academic","Other Academic","Other Academic",
        "Other Academic","Other Academic","Other Academic","Other Academic",
        "Other Academic","Other Academic"
    ]
}

df = pd.DataFrame(data)

vectorizer = TfidfVectorizer(stop_words="english")
X = vectorizer.fit_transform(df["Message"])
y = df["Label"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = MultinomialNB()
model.fit(X_train, y_train)
print(" Model trained!\n")

def extract_text_from_image(image_path):
    text = pytesseract.image_to_string(Image.open(image_path))
    text = re.sub(r'[\n\r]+' , ' ', text)
    text = re.sub(r'^\w\s:/,-,]', ' ', text)
    text = re.sub(r'\s+', ' ', text).strip()
    return text

def extract_datetime(message):
    date_patterns = [

```

```

        r'\d{1,2}(st|nd|rd|th)?\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)[\s,]*\d
{0,4}',
        r'(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)\s\d{1,2},?\s?\d{2,4}?',
        r'\d{1,2}/\d{1,2}/\d{2,4}',
        r'\d{4}-\d{1,2}-\d{1,2}',
        r'\d{1,2}\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|Dec)'
    ]

```

```

time_patterns = [
    r'\d{1,2}(:\d{2})?\s?(AM|PM)',
    r'\d{1,2}(:\d{2})'
]

```

```

date = "Not found"

```

```

time = "Not found"

```

```

for pattern in date_patterns:

```

```

    match = re.search(pattern, message, re.IGNORECASE)

```

```

    if match:

```

```

        date = match.group(0)

```

```

        break

```

```

for pattern in time_patterns:

```

```

    match = re.search(pattern, message, re.IGNORECASE)

```

```

    if match:

```

```

        time = match.group(0)

```

```

        break

```

```
return date, time
```

```
def extract_registration_deadline(message):
```

```
    pattern = r'((register|registration|audition|competition|deadline|last  
date)[^.,\n]*?(\d{1,2}(st|nd|rd|th)?\s?(Jan|Feb|Mar|Apr|May|Jun|Jul|Aug|Sep|Oct|Nov|De  
c)\s?\d{0,4})\d{1,2}/\d{1,2}/\d{2,4}))'
```

```
    match = re.search(pattern, message, re.IGNORECASE)
```

```
    return match.group(0) if match else "Not found"
```

```
def set_calendar_reminder(message, category, date, time):
```

```
    if category in ["Event", "Tech Talk", "Audition"]:
```

```
        print(f" Reminder set for {category}!")
```

```
        print(f" Event: {message}")
```

```
        print(f" Date: {date}")
```

```
        print(f" Time: {time}")
```

```
    else:
```

```
        print(f" No reminder for {category} messages")
```

```
def classify_message(message):
```

```
    X_new = vectorizer.transform([message])
```

```
    category = model.predict(X_new)[0]
```

```
    date, time = extract_datetime(message)
```

```
    registration = extract_registration_deadline(message)
```

```
    print("\nMessage:", message)
```

```
    print("Predicted Category:", category)
```

```
    print(" Event Date:", date)
```

```
    print(" Event Time:", time)
```

```
    print(" Registration / Deadline:", registration)
```

```
print("-"*60)
```

```
set_calendar_reminder(message, category, date, time)
```

```
print("-"*60)
```

```
def menu():
```

```
    while True:
```

```
        print("\n Academic Message Classifier")
```

```
        print("1. Test with default sample messages")
```

```
        print("2. Enter your own message")
```

```
        print("3. Test with OCR from an image")
```

```
        print("4. Exit")
```

```
        choice = input("Choose an option (1-4): ")
```

```
    if choice == "1":
```

```
        test_messages = [
```

```
            "Cyber Security Workshop on 15th Sep 2025 at 11:30 AM",
```

```
            "Submit assignment 3 by next Wednesday",
```

```
            "Music Fest happening on 18th Sep 2025 at 6 PM",
```

```
            "Auditions for College Drama on 15th Sep at 2 PM"
```

```
        ]
```

```
        for msg in test_messages:
```

```
            classify_message(msg)
```

```
    elif choice == "2":
```

```
        msg = input("Enter your message: ")
```

```
        classify_message(msg)
```

```

elif choice == "3":
    path = input("Enter image path: ")
    ocr_text = extract_text_from_image(path)
    print("\n OCR Extracted:", ocr_text)
    classify_message(ocr_text)

elif choice == "4":
    print(" Exiting. Goodbye!")
    break

else:
    print(" Invalid choice. Try again.")

if __name__ == "__main__":
    menu()

```

Input/Output Formats

- **Input:**
 - User-provided **text message** (e.g., *"Cyber Security Workshop on 15th Sep 2025 at 11:30 AM"*).
 - **Image file** containing text (processed through OCR).
- **Output:**
 - **Predicted Category** → Event / Tech Talk / Audition / Other Academic.
 - **Extracted Details** → Date, Time, and Registration/Deadline (if present).
 - **Reminder Simulation** → Generates reminder for Events, Tech Talks, and Auditions.

```

    ✓ Model trained!

    Academic Message Classifier
    1. Test with default sample messages
    2. Enter your own message
    3. Test with OCR from an image
    4. Exit
    Choose an option (1-4): 1

    Message: Cyber Security Workshop on 15th Sep 2025 at 11:30 AM
    Predicted Category: Tech Talk
    📅 Event Date: 15th Sep 2025
    🕒 Event Time: 11:30 AM
    📄 Registration / Deadline: Not found

    ⚠️ Reminder set for Tech Talk!
    Event: Cyber Security Workshop on 15th Sep 2025 at 11:30 AM
    📅 Date: 15th Sep 2025
    🕒 Time: 11:30 AM

    -----

    Message: Submit assignment 3 by next Wednesday
    Predicted Category: Other Academic
    📅 Event Date: Not found
    🕒 Event Time: Not found
    📄 Registration / Deadline: Not found

    ⚠️ No reminder for Other Academic messages

    -----

    Message: Music Fest happening on 18th Sep 2025 at 6 PM
    Predicted Category: Audition
    📅 Event Date: 18th Sep 2025
    🕒 Event Time: 6 PM
    📄 Registration / Deadline: Not found

    ⚠️ Reminder set for Audition!
    Event: Music Fest happening on 18th Sep 2025 at 6 PM
    📅 Date: 18th Sep 2025
    🕒 Time: 6 PM

    -----

    Message: Auditions for College Drama on 15th Sep at 2 PM
    Predicted Category: Audition
    📅 Event Date: 15th Sep
    🕒 Event Time: 2 PM
    📄 Registration / Deadline: Auditions for College Drama on 15th Sep

    ⚠️ Reminder set for Audition!
    Event: Auditions for College Drama on 15th Sep at 2 PM
    📅 Date: 15th Sep
    🕒 Time: 2 PM

    -----

    Academic Message Classifier
    1. Test with default sample messages
    2. Enter your own message
    3. Test with OCR from an image
    4. Exit
    Choose an option (1-4): 2
    Enter your message: Darpan dance audition at 6pm in Amoga hall on 5th sep.

    Message: Darpan dance audition at 6pm in Amoga hall on 5th sep.
    Predicted Category: Audition
    📅 Event Date: 5th sep
    🕒 Event Time: 6pm
    📄 Registration / Deadline: audition at 6pm in Amoga hall on 5th sep

    ⚠️ Reminder set for Audition!
    Event: Darpan dance audition at 6pm in Amoga hall on 5th sep.
    📅 Date: 5th sep
    🕒 Time: 6pm

    -----

    Academic Message Classifier
    1. Test with default sample messages
    2. Enter your own message
    3. Test with OCR from an image
    4. Exit
    Choose an option (1-4): 3
    Enter image path: /content/WhatsApp Image 2025-08-23 at 20.48.28.jpeg

    📄 OCR Extracted: AE ge :5 ee othr. aie, Bvcevt get Bice 08: ji i com GLB INNOVATI

```

```

Message: AE ge :5 ee othr. aie, Bvcevt get Bice OB: ji i com GLB INNOVATION CELL Lau
Predicted Category: Tech Talk
📅 Event Date: 25th Aug
🕒 Event Time: 11:30 AM
📄 Registration / Deadline: Not found
-----
🔔 Reminder set for Tech Talk!
Event: AE ge :5 ee othr. aie, Bvcevt get Bice OB: ji i com GLB INNOVATION CELL Lau
📅 Date: 25th Aug
🕒 Time: 11:30 AM
-----

📁 Academic Message Classifier
1. Test with default sample messages
2. Enter your own message
3. Test with OCR from an image
4. Exit
Choose an option (1-4): /content/WhatsApp Image 2025-08-23 at 20.48.28.jpeg
⚠️ Invalid choice. Try again.

📁 Academic Message Classifier
1. Test with default sample messages
2. Enter your own message
3. Test with OCR from an image
4. Exit
Choose an option (1-4): 3
Enter image path: /content/WhatsApp Image 2025-08-23 at 20.48.28.jpeg

📄 OCR Extracted: AE ge :5 ee othr. aie, Bvcevt get Bice OB: ji i com GLB INNOVATION

Message: AE ge :5 ee othr. aie, Bvcevt get Bice OB: ji i com GLB INNOVATION CELL Lau
Predicted Category: Tech Talk
📅 Event Date: 25th Aug
🕒 Event Time: 11:30 AM
📄 Registration / Deadline: Not found
-----
🔔 Reminder set for Tech Talk!
Event: AE ge :5 ee othr. aie, Bvcevt get Bice OB: ji i com GLB INNOVATION CELL Lau
📅 Date: 25th Aug
🕒 Time: 11:30 AM

```

```

📁 Academic Message Classifier
1. Test with default sample messages
2. Enter your own message
3. Test with OCR from an image
4. Exit
Choose an option (1-4): 3
Enter image path: /content/WhatsApp Image 2025-09-12 at 12.25.21.jpeg

📄 OCR Extracted: oferta cellists ss Cer Unmask aT ug Secure the future. Step into the TCS iieadiSsstode Season 10 battleground TCSHackQuest

Message: oferta cellists ss Cer Unmask aT ug Secure the future. Step into the TCS iieadiSsstode Season 10 battleground TCSHackQuest
Predicted Category: Event
📅 Event Date: Not found
🕒 Event Time: Not found
📄 Registration / Deadline: Not found
-----
🔔 Reminder set for Event!
Event: oferta cellists ss Cer Unmask aT ug Secure the future. Step into the TCS iieadiSsstode Season 10 battleground TCSHackQuest
📅 Date: Not found
🕒 Time: Not found
-----

📁 Academic Message Classifier
1. Test with default sample messages
2. Enter your own message
3. Test with OCR from an image
4. Exit
Choose an option (1-4): 4
👋 Exiting. Goodbye!

```


7. Results and Discussion

Sample Results (from screenshots)

1. **Input:** *“Cyber Security Workshop on 15th Sep 2025 at 11:30 AM”*
 - Predicted Category: **Tech Talk**
 - Extracted Date: 15th Sep 2025
 - Extracted Time: 11:30 AM
 - Reminder: Set
2. **Input:** *“Submit assignment 3 by next Wednesday”*
 - Predicted Category: **Other Academic**
 - Extracted Date: Not found
 - Reminder: Not applicable
3. **Input:** *“Music Fest happening on 18th Sep 2025 at 6 PM”*
 - Predicted Category: **Audition**
 - Extracted Date: 18th Sep 2025
 - Extracted Time: 6 PM
 - Reminder: Set
4. **OCR Test:** Extracted noisy text still classified correctly as **Tech Talk**, proving robustness of the classifier.

Error Analysis

- OCR sometimes introduced noisy text, but classifier still predicted correct category.
- Regex extraction missed vague references like *“next Wednesday”*.

8. Conclusion

The project successfully built a lightweight Academic Message Classifier using NLP. It categorized academic notices, extracted key details, and simulated reminders. OCR integration made it adaptable for scanned announcements.

Limitations

- Small dataset (50 samples).
- Regex cannot handle vague temporal expressions (e.g., “*next week*”).

Future Scope

- Expand dataset with hundreds of real student messages.
- Use contextual models like **BERT** for improved classification.
- Integrate with Google Calendar for real event reminders.

9. References

1. Scikit-learn Documentation – <https://scikit-learn.org/>
2. Pytesseract OCR – <https://pypi.org/project/pytesseract/>